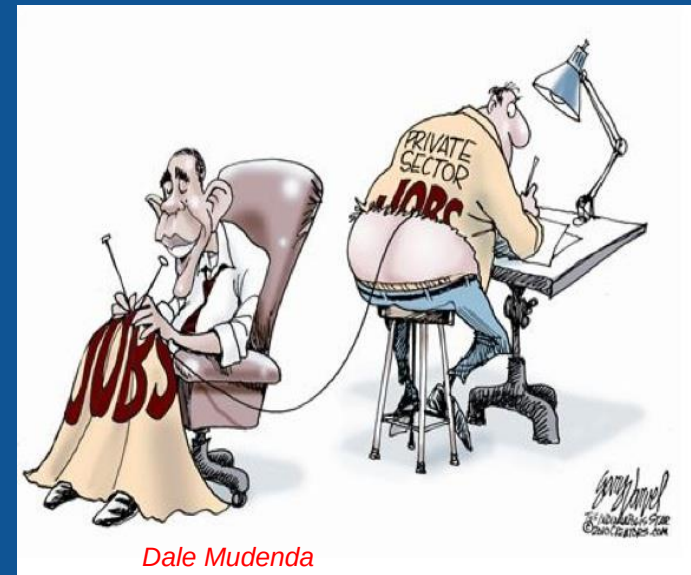




Government Sector

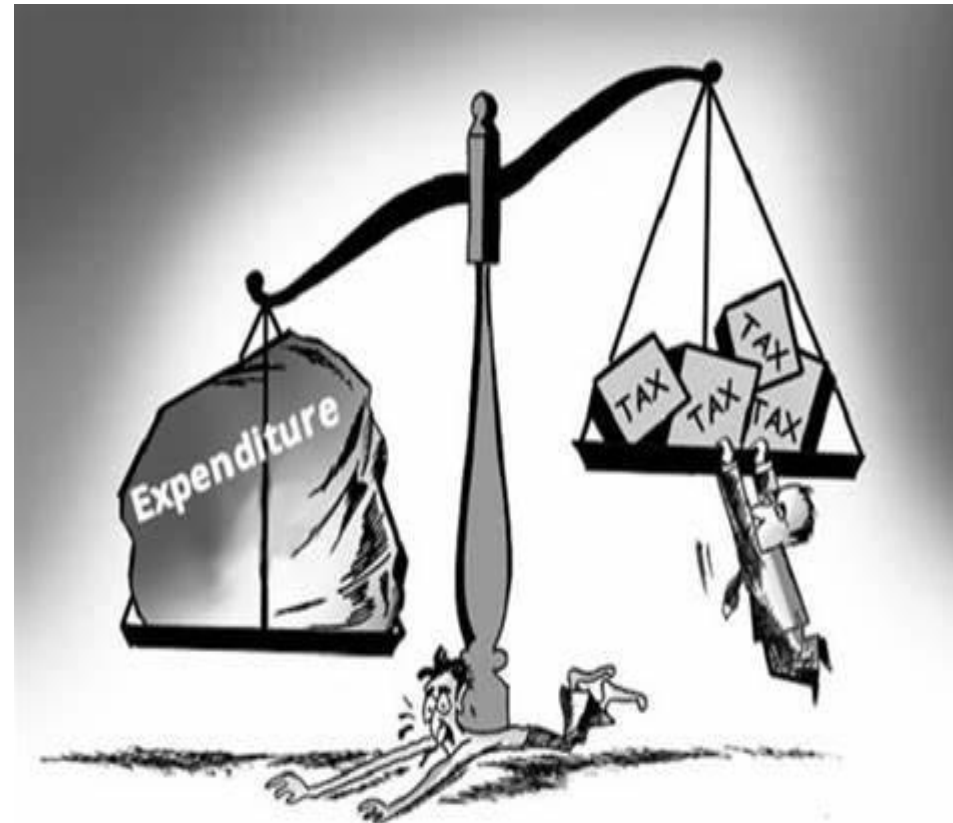
Lecture 8: Macroeconomics and the Budget





Lecture Outline

- **Government expenditure**
- **Taxation**
- **The budget deficit**
- **Government debt**
- **Debt Management**
- **Fiscal Policy in General**





Fiscal Policy

- ❖ In this lecture we will create a framework for thinking about how **taxation** and **government** spending affect the national economy based on the aggregate demand and supply model we saw last week
- ❖ *Expenditure in a closed economy and Open economy*

$$Y = C + I + G + (X - M)$$

$$❖ Y = C + S + T$$

- ❖ *Y – Expenditure/Income [GDP]*
 - C – Consumption*
 - I – Investment*
 - G – Government spending*
 - S – Saving*



Basic Economic Equilibrium model

Remember – leakages equal injections.....

$$Y = GDP = C + I + G = C + S + T + (X - M)$$

or

$$G - T = (S - I) + (X - M)$$

BUDGET
DEFICIT/
DEBT
FINANCING

BALANCE OF
PAYMENTS GAP

SAVINGS GAP

Y – Expenditure/Income [GDP]

C – Consumption

I – Investment

G – Government spending

S – Saving

T – Taxes

X – Exports

M – Imports



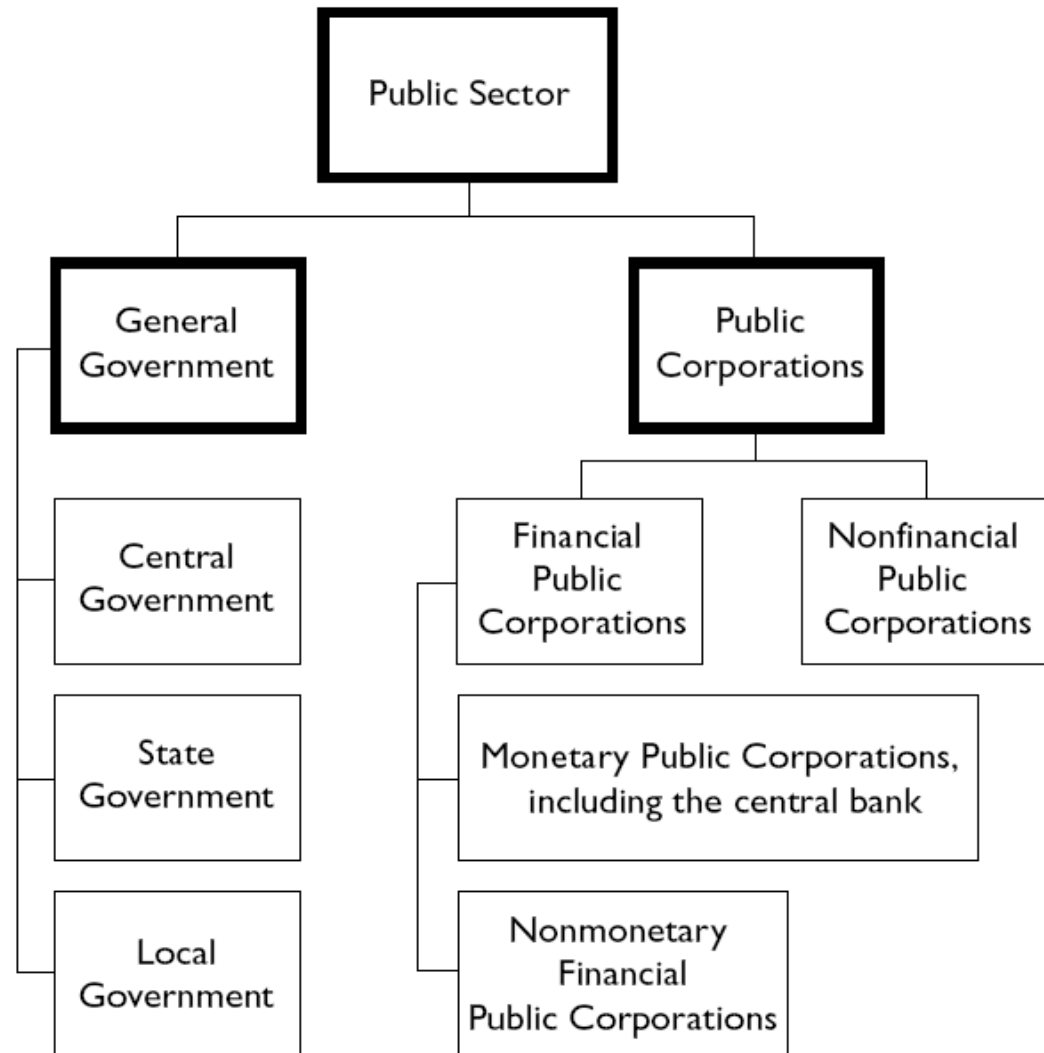
Fiscal Policy

- Fiscal policy refers to government's decisions about the level of taxation and public spending
- It is more than just simple budgeting - because choices about how much to spend and how to raise the necessary funds can have dramatic effects on the economy
- Recall: **$Y = C + I + G + NX$**
 - Fiscal policy affects the economy by increasing or decreasing aggregate demand
 - Shifts in AD translate into either higher or lower output and price levels
 - Fiscal policy affects AD through two channels
 - a) Increase in government spending
 - b) Taxations policy



Government Spending

- ❑ Government spending-is current government spending on goods and services
- ❑ Total government spending however also includes government capital formations -(like roads) and transfers (**include debt interest**)
- ❑ **Implicitly-** we include government investment along with private investment
- ❑ Transfers are often not modelled because they turn up in disposable income



Source: Government Finance Statistics (GFS). IMF



Government Budget

- The government **budget** describes what goods and services the government will buy during the coming year, what transfer payments it will make and how it will pay for them
- A budget is the spending and revenue plan of an individual, a firm or a government
- *Most of its spending is financed by taxes. When spending exceeds taxes, there is a budget deficit*
- *When taxes exceed spending, there is a budget surplus*



Statement of Government Operations

Tableau des opérations financières de l'Etat (TOFE)	
Revenues	
Revenues and grants	T
Total Revenues	T1
Grants	T2
Expenditure	
Total expenditure	G
Recurrent expenditure	G1
Wages and salaries	G11
Purchases of goods and services	G12
Transfers	G13
Interest payments	G14
Capital expenditure	G2
Balance of government operations	
Overall balance	S = T - G
"The Line"	
Financing	F = F1 + F2
Domestic financing	F1
External financing	F2



Zambian Budget -2018 outcomes (Kwacha billions)

Revenues	Target	Outturn
Domestic	21.05	20.08
grants	1.115	0.27
Financing	7.505	7.944
o/w foreign	4.98	1.304
o/w domestic	2.5	6.6
Total	29.67	28.29
Expenditure	Target	Outturn
PES	10.01	10.58
Purchase of Government goods and services	2.85	2.21
Interest payments	4.18	4.59
Grants and other payments	3.96	4.18
Social benefits	1.264	1.099
Other expenses	0.346	0.329
Assets	6.049	2.607
Liabilities	1.269	1.005
Total Expenditure	29.94	26.6



The 'Government Fiscal Constraint'

$$\square G^K + G^R + r(D) = T + \text{Grants} + \Delta(D)$$

G^K – Capital Expenditure

G^R – Recurrent Expenditure

$r(D)$ – Interest payments on Debt

T – Domestic Resources (Tax and Non-Tax)

$\Delta(D)$ – Borrowing (change in the stock of Debt)

We will get back to this





Taxation

- ❖ *Government spending is having to be financed*
- *Taxes – are compulsory transfers of money from private individuals, groups or institutions to the government*
- *Overall fiscal policy includes changes in G and T designed to achieve full employment and stable prices in the economy*
- *Generally, government action aimed at keeping AD and output close to potential output is called **Stabilization policy***
- **Can be undertaken through**
 - Government spending
 - **Taxation**



Fiscal Policy

- Government spending generally shifts the AD to the right
- Government spending also has indirect effect on consumption and Investment through the multiplier
- Government also uses tax policy to drive AD
- Taxes act on AD directly through consumption and indirectly through investment etc.
- There are two types of fiscal policy
- *Expansionary fiscal policy* - that is decisions about government spending and taxation intended to increase AD
 - Increase in G and lower taxes both have expansionary effect on output
 - AD shifts to the right



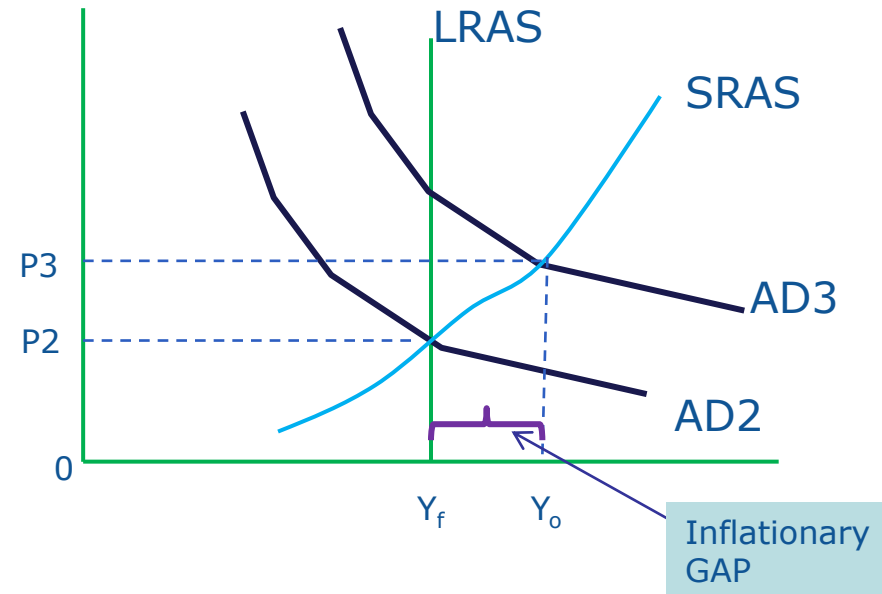
Fiscal Policy

- **Contractionary fiscal policy** – refer to the overall effects of decisions about government spending and consumption intended to decrease AD
 - Decreased G and higher T both have contractionary effects, shifting AD to the right
 - Governments also tend to change both taxes and its spending at the same time
 - If government spending is offset by a rise in tax rate such that tax revenue changes exactly match the change in government spending
 - In equilibrium the planned leakages will exactly equal injections
 - **$I+G+X = S+T+M$**



Fiscal policy

- Contractionary or restrictive fiscal policy can help to address the inflationary gap caused by excessive growth of AD
- The options:
 - ❑ **Decrease government spending**
 - ❑ **Raise taxes**
 - ❑ **Use a combination of these two policies**
- Example: if the economy is operating at AD3 real output exceeds the full potential. This creates an inflationary gap
- We can reduce the inflationary gap by reducing government spending so that AD3 shifts to AD2 – full employment level
- An option – government can use tax increase to reduce consumption



Or investment spending
Investment spending reduces consumption or investment spending , reducing AD from AD3 to AD2



Fiscal Policy

- A balanced increase in planned govt spending and net tax revenue must increase equilibrium output
- The balanced budget multiplier says that a rise in government spending plus an equal rise in taxes leads to higher output

- Assume our model

- $Y = C + I + G + NX$
- $C = c(Y - T)$

Obtain the multiplier

- $Y = c(Y - T) + I + G + NX$
- $(1 - c)Y = I + G - cT + NX$
- Make y Subject of the formulae

$$Y = \frac{1}{1-c} (I + G - cT + NX)$$

If you increased government spending, we obtain:

$$\Delta Y = \frac{1}{1-c} (\Delta I + \Delta G - c\Delta T + \Delta NX)$$

Holding all else constant an increase in government spending increases income by the multiplier $\frac{1}{1-c}$

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1-c}$$



Balanced budget multiplier

- Now assume that government runs a balanced budget where a rise in government spending is matched by an equal rise in taxes
- In this case **$G = T$**
- Recall $\Delta Y = \frac{1}{1-c} (\Delta I + \Delta G - c\Delta T + \Delta NX)$
- Insert G where there is T in the model:

$$\Delta Y = \frac{1}{1-c} (\Delta I + \Delta G - c\Delta G + \Delta NX)$$

Replace T
by G

$$\Delta Y = \frac{1}{1-c} (\Delta I + (1-c)\Delta G + \Delta NX)$$

The balanced budget multiplier is :

$$\frac{\Delta Y}{\Delta G} = \frac{1-c}{1-c} = 1$$

Government has to have the political courage to raise tax revenue in line with higher expenditure



The **budget multiplier** says that a rise in government spending plus an equal rise in taxes leads to higher output by the value of the increase in G



Budget Deficits and Surpluses

- Governments can also spend more than they raise from taxes
 - **$G > T$.**
 - This means government has a deficit budget
 - The budget deficit is the excess of government spending over government receipts during a particular period.
 - It represents the amount of money govt spends beyond the revenue
 - When government decides to spend more , we say it is running an **expansionary fiscal policy**
- It is also possible for govt to have a **budget surplus** which is an amount of revenue a government brings in beyond what it spends
 - **$G < T$**
 - In this case, government is saving.



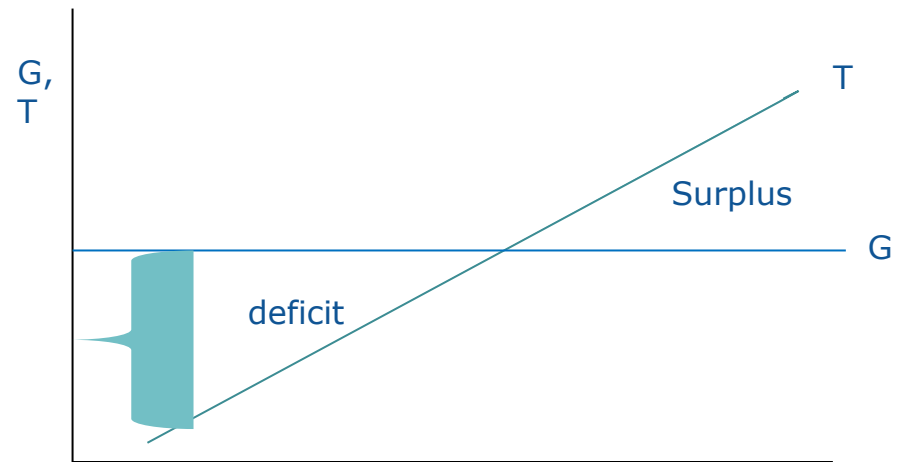
Built in Stabilisers

- To an extent government tax revenues change automatically over the course of the business cycles in ways that stabilize the economy
- This automatic adjustment or built in stability is part of the non-discretionary budgetary policy
- This more likely when the tax is not charged as lumpsum but as a function of tax rate so that the multiplier becomes affected as follows:
 - $Y = C + I + G + NX$
 - $C = c(Y - T)$ and $T = tY$
 - $Y = c(Y - tY) + I + G + NX$
 - **Make y subject**
 - $(1 - (1 - t)c)y = I + G + NX$
 - $Y = \frac{1}{1 - (1 - t)c} (I + G + NX)$
- ❖ The multiplier now has $(1 - t)$ shows the amount by which disposable income is less than national income
- ❖ higher tax rate t reduces the multiplier
- ❖ However, the tax yields more tax revenue as GDP rises



Built in Stabilisers

- In particular personal income taxes have progressive rates
- They tend to generate more than proportionate increases in tax revenue as GDP expands
- **An in-built or automatic stabilizer** – is a mechanism that increases government's budget deficit or reduces the surplus during recessions and increases government budget surplus during booms without any action by policy makers



Automatic stabilisers have economic importance:

- 1. Taxes reduce spending and aggregate demand*
- 2. Reductions in spending are desirable when the economy is booming while increases in spending are desirable then the economy is slumping*



- ❖ All leakages are automatic stabilizer
 - ❖ A higher saving rate (hence a lower marginal propensity to consume) reduces the multiplier
 - ❖ Government can still use discretionary fiscal policy
- Discretionary fiscal policy is decisions about tax rates and levels of government spending

- ❖ When other components of AD are abnormally low, the government can boost demand by cutting taxes, raising spending, or both

The Limits To Fiscal Policy

- ❖ *There are several problems of timing that arise in connection with fiscal policy*



Limits to Fiscal Policy

1. *Recognition lag – this is the time between the beginning of the recessions or inflation and the certain awareness that it has actually happened*
 - It arises because of the difficulty in predicting the future course of economic activity
 - Forecasting tools may provide an indicator of the direction of the economy, the economy may be already in a recession
 - The situation could be worse than if it was identified and acted on sooner



Limits to Fiscal Policy

2. *Administrative lag*– the wheels of democratic govt turn slowly. Often, there is a significant lag between the time the need for fiscal policy action is recognized and the time action is taken
 - Parliament has to approve revenue and expenditure changes
 - If its not in session the changes will need to wait for it to reconvene
 - By the time it reconvenes, the economic situation may have changed rendering the belated policy action inappropriate
3. *Operational lag* – there is also a lag between the time fiscal action is taken and the time that policy action affects output, employment and or the price level
 - **Even** if fiscal policy is quickly approved, it takes time for govt spending to be implemented.



Evaluating Fiscal Policy

- ❖ How do we know if fiscal policy is expansionary, neutral or contractionary in a given period?
- ❖ In general, the deficits and surpluses may not be appropriate because they may reflect some automatic adjustment or changes in tax revenue due to changes in GDP
- ❖ To evaluate the status of fiscal policy, we need to:
 - 1. Adjust fiscal deficits to eliminate automatic changes in tax revenue**
 - 2. Compare the size of the adjusted budget deficits/surpluses to GDP levels**

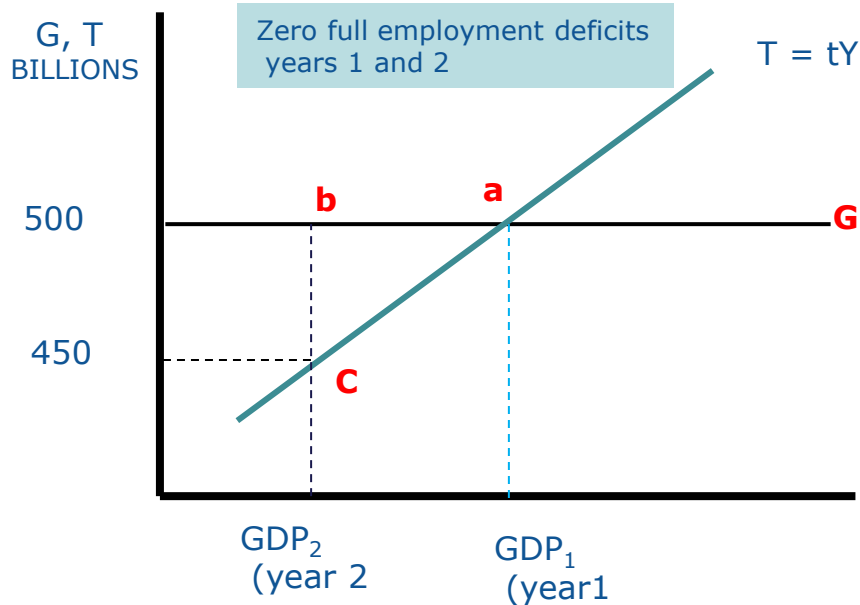


Structural/Cyclically adjusted Budget

- ❖ The structural budget/full-employment budget shows what the budget will be if output is at potential output
- ❖ The cyclically adjusted budget is used to adjust the actual budget deficits/surpluses to eliminate the automatic changes in tax revenue
 - It measures what the actual budget deficit or surplus would be with existing tax rates and govt spending levels if the economy had achieved full employment
 - We compare actual govt spending for each year with the tax revenue that would have occurred in that year if the economy had achieved full employment GDP
 - The cyclically adjusted budget is expressed as $G - tY^*$
 - It is affected by proactive fiscal decisions



Structural/Cyclically adjusted Budget



- Now if a recession occurred, output fell to GDP₂ in year 2.
- If Govt doesn't take discretionary action, tax revenue falls to K450 bn at point c while government spending remains at K500 billion at b
- We have a budget deficit of:
 - **G – T = deficit**
 - **500 – 450 = 50**
- The cyclical deficit here is just a by-product of the economy's sliding into a recession and not policy
- Government has not changed its fiscal policy stance

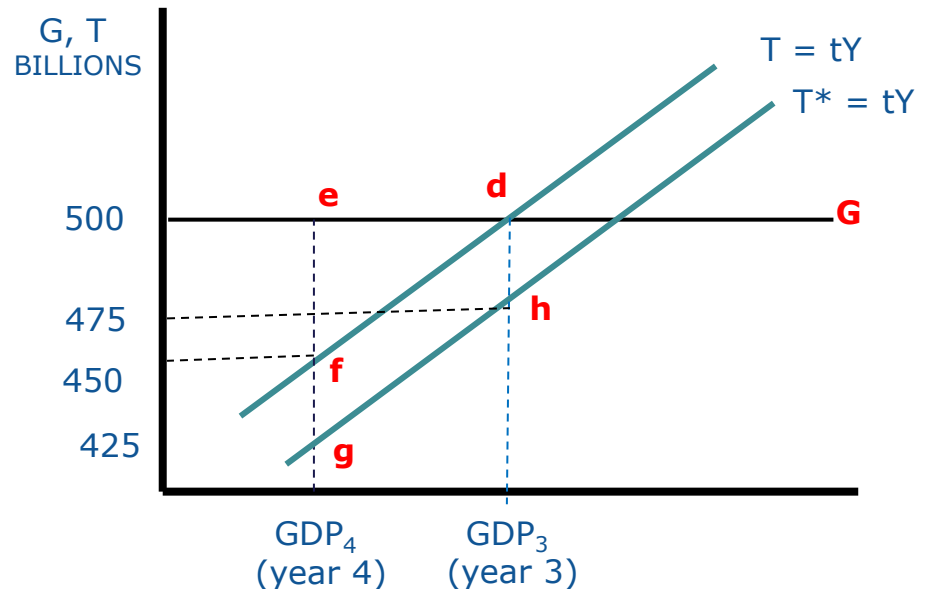
- ❖ At point a in year 1, full employment expenditure is K500 Bn and tax revenue equal K500 Bn
- ❖ The cyclically adjusted budget is zero
- ❖ The full employment deficit as percentage of GDP is zero



Structural/Cyclically adjusted Budget

- Now suppose output fell from GDP3 to GDP4 and
- Govt implements a tax rate reduction that shifts the curve from T to T^*
- What has happened to the cyclical adjusted deficit?
- Govt expenditure in year 4 is K500 Bn at e
- Compare with the tax revenue of K475 billion that would occur if the economy achieved its full employment GDP

Zero full employment deficits
years 3; K25 bn full employment deficit year 4





Public Debt and deficits

- Recall: A deficit –is the amount by which government expenditure exceeds its revenue in a particular year.
- The public debt – is closely related to the deficits
- The public debt is the total accumulation of the government's total deficits and surpluses that have occurred over time
- Debts is the total amount that government owes
- The ***national debt*** is the stock of government debt outstanding
- How does government go into debt?
- Government goes into debt through many ways
- First, it borrows money from people by **selling treasury securities**
 - The idea is: government accepts money from people with an obligation to pay them back at a certain date
 - Some treasuries securities are short term (90 days, 180 days, 270 days etc but less than a year .) and other long term 1 year or even 30 years



Public Debt and deficits

- **Short-term securities** ones are called treasury bills or T-bills
- They are loans than mature in less than a year
- Government promises the buyers to pay them a certain amount on a fixed date
- People bid for T-bills when they are issue
- **Long term securities** are called Treasury Notes or bonds
- The purchaser receives periodic (every 6 months) interest payment
- The return may be higher than on T-bills

Treasury notes could be 2 to 7 years

Bonds can mature even after 30 years

Euro bond – is a bond denominated in foreign currency

Risks to holding gov't securities

- **Inflation – may eat away** their value
- Some countries, hedge against inflation by offering Treasury Inflation Protected Securities (TIPS)
- The original amount of investment is pegged to changes in overall inflation while the interest rate remains fixed



Government Debt

Zambia's Developments in Fiscal sector (2014-2021)

Year	Total Government Expenditure (% GDP)	Government Deficit (% GDP)	General Government Revenue (% GDP)	External Debt Stock (% GDP)	Domestic Debt (% GDP)
2014	24.7%	-6.6%	18.1%	34.6%	17.0%
2015	28.3%	-9.7%	18.6%	56.5%	15.3%
2016	23.9%	-5.9%	18.0%	74.9%	12.0%
2017	25.0%	-7.7%	17.3%	92.8%	19.6%
2018	27.7%	-8.5%	19.2%	90.8%	21.2%
2019	29.8%	-9.7%	20.1%	119.0%	26.7%
2020	34.1%	-14.3%	19.8%	151.6%	39.2%
2021	31.6%	-9.0%	22.6%	124.7%	45.5%



Public Debt

- Treasury securities are purchased as financial investment by:
 - individuals
 - governments and
 - Commercial banks (both local and abroad)
- WHY HOLD GOVT DEBT
- They are relatively safe investments especially to other investments
 - Real estates, stocks can easily collapse
 - Government, though can refuse to pay rarely do so
- **External debt:** is debt owed to agents located abroad, that are the holders of the debt are not resident in the economy that issued the debt.
 - Govt pays interest periodically
 - This must be financed by either further borrowing or by taxation
- **Internal debt:** is held by residents of the economy in which it is issued.
 - The government borrows off its citizens by providing bonds which compete with private capital



Composition of Zambian debt

Year	2017	2018	2019	2020	2021	2022
Multilateral Debt (US\$ billion)	1.6	1.9	1.9	2.0	2.1	
Bilateral Debt (US\$ billion)	0.4	0.3	0.3	0.3	0.3	
Export & Suppliers (US\$ billion)	2.2	2.5	2.7	2.8	2.9	
Private Banks (US\$ billion)	4.6	5.3	5.3	7.6	7.6	
Guarantees (US\$ billion)	0.7	1.2	1.2		1.8	
Total External Debt (US\$ billion)	9.5	11.2	11.4	12.7	14.71	
Domestic Debt (K billion)		58.26	80.2	130.2	180.23	



Causes of debt crisis in developing countries

○ **External factors**

- External borrowing to compensate for rising import prices and slumps in prices of their principal export commodities.
- Imprudent lending on the part of commercial banks in a period of petrodollar glut.
- Sharp rises in interest rates and the value of the dollar that can exacerbated the debt situation, increasing the cost of debt service.

○ **Internal Factors**

- a. Lack of restraint on the part of governments and borrow to cover budget deficits.
- b. **Unwise investments** i.e. unproductive investments (White elephants)
- c. **High levels of consumption** in relation to the country's resources
- d. During **calamities** such as wars, earthquakes, floods, famine
- e. Borrow to accelerate economic growth and development



Benefits of Debt

- ❖ Allows govt to be flexible when something unexpected happens – e.g.. floods and droughts etc.
- ❖ It can pay for investments that will lead to economic growth and prosperity in the long run

❑ **Costs of debt**

1. Direct costs – interest government pays to people it has borrowed from. These payments are substantial and can add a huge strain to the treasury
 - The burden depends on interest rate. Higher interest rates, like on commercial loans means govt has to pay more
 - The rate return also depends on investors' confidence in govt' ability to pay back its debt
 - If govt is poorly rated, interest will be higher otherwise lower
 - The self-reinforcing spiral of investor doubt and a higher interest rate has in the past caused Zambia default



Costs of debt

- 2. Indirect Cost:** of govt debt – it can distort the credit markets and slow economic growth
- Government borrowing can crowd out private borrowing. When government borrows it increases the demand for credit and increases the price of credit in the economy
 - A higher interest rate increases the cost of borrowing for businesses that want to invest
 - Finally, who bears the burden of government debt? The people today benefit from debt to spend on services but people tomorrow, future generations, will have to pay the debt
 - The cost of services are passed on to future generations



Debt redemption

1. Repudiate the debt i.e., refuse to pay. The problem is that the government will find it difficult to borrow in the future at a reasonable rate
2. Establish a sinking fund: The government can regularly save for the retirement of the debt and uses the funds for this purpose when it has accumulated enough
3. Regularly retire a small portion of the debt.
4. Pay off short term debt with corresponding surplus.
5. Restructure the debt/Refund: convert the existing debt into a new one of longer maturity i.e., holders purchase a new loan.
6. Debt conversion: a high interest public debt is converted into low interest public debt. Thus the government contracts a new debt at low - interest cost debt and uses



8. Ask for debt relief: Lowers credit rating of the country.
9. Debt cancellation: Lowers credit rating of the country.
10. Debt rescheduling; Lowers credit rating of the country.
11. Compulsory reduction in the interest rate. Happens when a government is confronted with a financial crisis



macro-fiscal stability

Aggregate fiscal discipline is paramount...

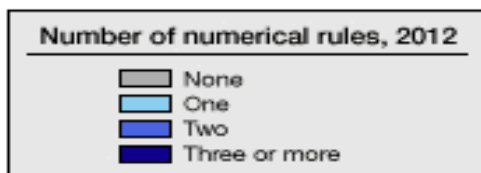
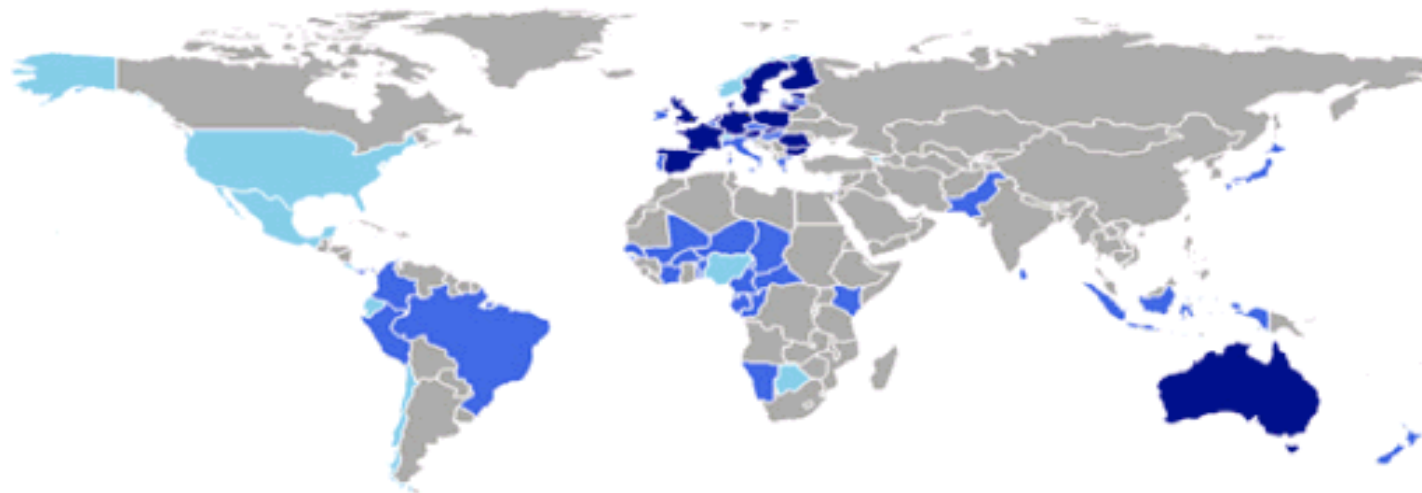
- ✓ Adequate systems; macro-fiscal policy, medium term budget planning et.al. must be robust to prevent uncontrolled 'additional' financing over and above what is planned in the Budget;
- ✓ Fiscal rules; 'bind' a government (central and/or sub-national) into a specific fiscal behaviour and prevent discretionary intervention – examples:
 - Capping deficit/GDP and Debt/GDP ratios in legislation;
 - Precluding local government from incurring debt;
 - 'Golden rule' debt financing only for capital spending.



PFM and macro-fiscal stability

Rules for: Debt, Deficit, Revenue, Expenditure

Countries with Fiscal Rules (National and Supranational), 2012



Source: National authorities; and IMF staff assessment.
Note: Based on fiscal rules in effect by end-March 2012.



Key message...

...how the Budget is financed matters!

